

— INDEPENDENT CASE STUDY · PREPARED FOR KOLIBRI GAMES

Idle Bank Tycoon — Economy Design Case Study

Two economy levers, a late-game cash sink and rewarded-ad tuning, each built into a working, parameter-driven model with an A/B test plan.

+15.5% blended ARPDAU (modelled)	+41.7% rewarded-ad revenue, same audience	≈ \$2M incremental / yr at 200k DAU
400+ live formulas, one source-of-truth tab	~73 rewarded videos / user, among the genre's highest	2 levers, each run as a guard-railed A/B test

The opportunity is in the late game

Idle Bank Tycoon is a commercial success: third in revenue across Kolibri's portfolio, strong D1 retention **~48%**, and a young, ad-tolerant audience. Its growth constraint is **late-game depth and monetisation efficiency**, not acquisition. I modelled two levers, each one end to end.

OPPORTUNITY	THE LEVER	MODELLED OUTCOME
A. Loan Desk	A soft-currency sink: lock idle cash for a fixed term in exchange for a temporary income boost. Hard-currency early-repay and slots.	+\$0.009 ARPDAU (sink alone)
B. Ad tuning	Scale the rewarded reward to the player's income, lift opt-in views/day, and resequence interstitials. No extra ad load.	+41.7% RV rev. +\$0.019 ARPDAU

At **200k** DAU the two levers lift blended ARPDAU from **\$0.180** to **\$0.208**, on the order of **\$2M incremental annual revenue** at static DAU. The Loan Desk is also built to lift D7 by giving late-game players a decision to make. The two are complementary by segment: ads monetise the non-paying majority, the hard-currency sink serves the spending minority. No pay-to-win.

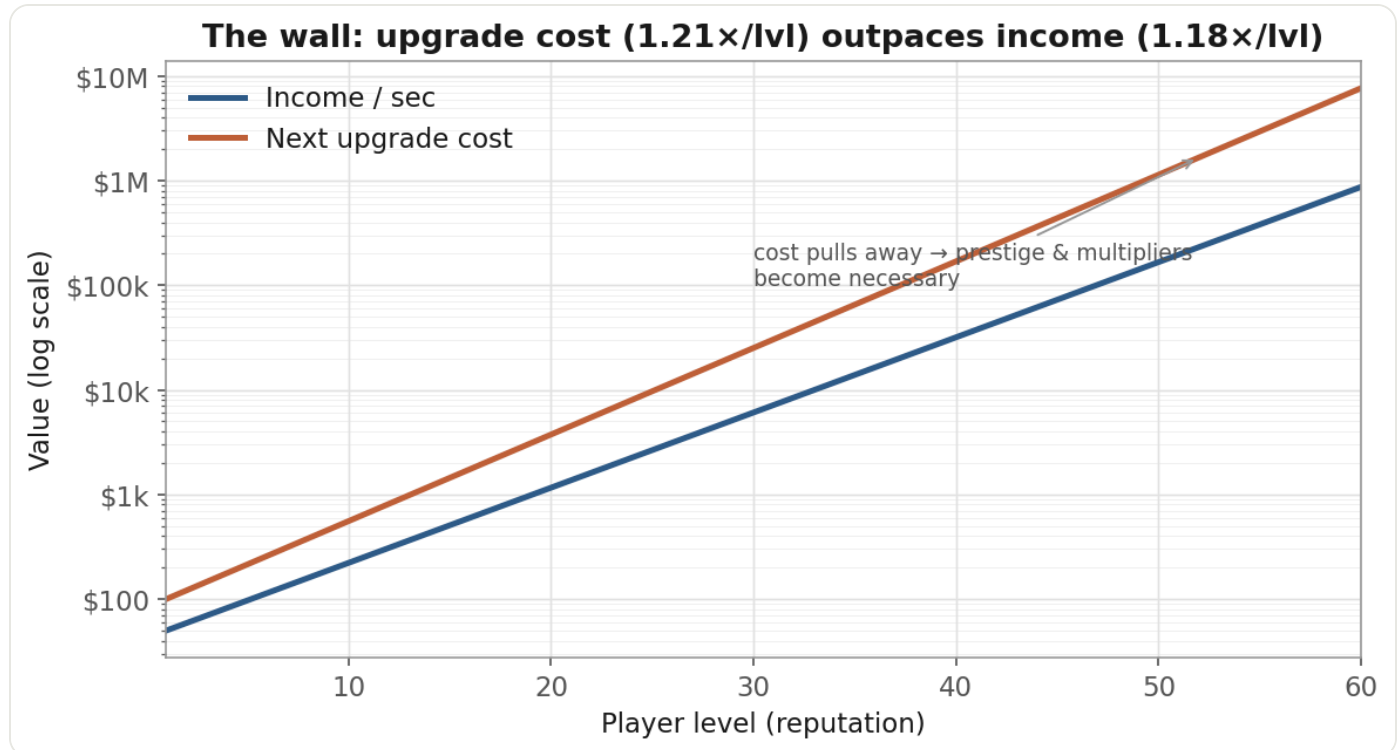
THE CONTEXT

What shapes the work

- **It is ad-led, not whale-led.** AppQuantum estimates US IAP revenue-per-download at **~\$0.6**, with 51% of users male and under 25. Rewarded video carries the revenue.
- **Retention is already strong** (D1 ~48%), so the opportunity is in the mid-to-late funnel and revenue-per-user, not the top of the funnel.
- **The late game lacks sinks.** AppQuantum's deconstruction recommends meta-depth and alternative soft-currency uses. Players reach a point where they hold far more cash than anything available costs.

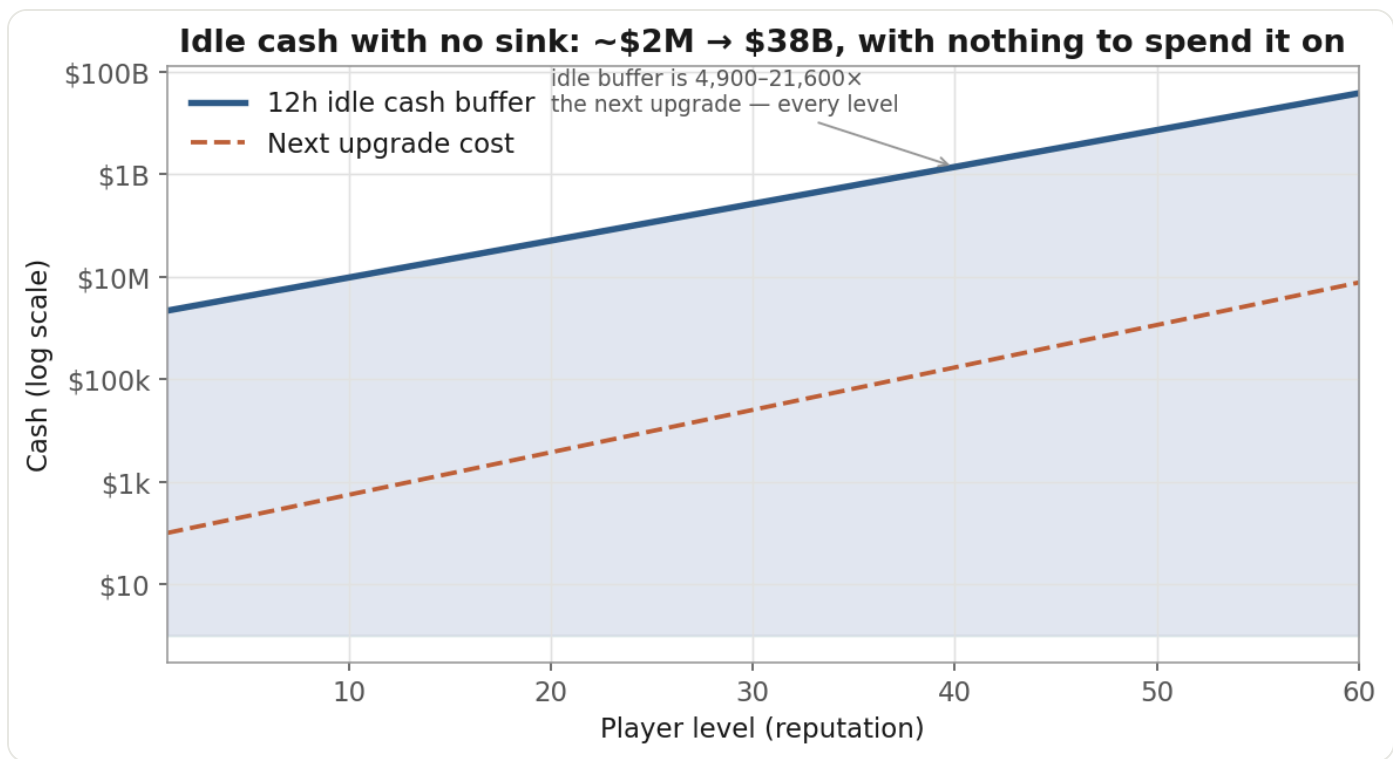
Diagnosing the cash-surplus problem

An idle economy runs on two curves. Cost rises exponentially $1.21 \times /lv1$; income rises more slowly $1.18 \times /lv1$. Cost always overtakes income, which is what makes multipliers and prestige necessary. Production is classically linear in units owned; here it is modelled per level, so both curves read as straight lines on the log axis below.



Cost pulls away from income, the wall that makes sinks and prestige necessary.

The 12-hour offline cap lets cash accumulate *between* the few upgrades that matter. Bank the offline buffer at each level's income rate against the next upgrade's cost, and the idle buffer sits at **4,900–21,600× the next upgrade at every level**, growing from about \$2M to about \$38B in absolute terms.



Idle cash with no sink: the surplus keeps growing while there is little to spend it on.

Opportunity A locks the surplus into timed loans. Opportunity B converts the same engagement into more rewarded-ad revenue, without changing the core balance.

The Loan Desk: a soft-currency sink

The concept is native to a bank. The player issues a **loan** by locking a share of idle cash for a fixed real-time term. While the loan runs, income/sec is boosted, so the player is earning interest. At maturity the principal returns with a small soft bonus. Players spend **Gold Ingots** to end a loan early or unlock extra slots, which is the hard-currency sink.

WORKED EXAMPLE: ~LEVEL-30 PLAYER	VALUE
Income / sec	~\$6,075
12h idle buffer (the surplus available)	~\$262M
Loan principal (40% of idle cash)	~\$105M
Income boost (1.5× for 4h)	\$6,075 → \$9,113/s
Extra cash over the 4h term	~\$43.7M
Boost reward as % of normal 4h earnings	50.0%
Early-repay fee (hard-currency sink)	25 ingots

SIX GUARDRAILS: ANTI-EXPLOIT, ANTI-INFLATION

- 1 Principal is a percentage of the live balance, so it grows with the player and resists hoarding.
- 2 The reward is time-based flow, not raw power. Prestige math is untouched and there is no pay-to-win.
- 3 The boost reward stays modest (50% of normal 4h), so loans complement active upgrading instead of replacing it. This is what prevents cannibalisation.
- 4 Every accelerator costs hard currency. None of them prints soft currency.
- 5 Free slots are limited, so the feature cannot dominate a session. Extra slots are a paid sink.
- 6 Soft interest is low, so the Loan Desk drains cash on net.

The goal is a feature players experience as generous that still drains more currency than it grants.

OPPORTUNITY B

Rewarded-ad reward tuning

The audience is young, low-spend, and ad-tolerant. Idle games serve about **73** rewarded videos per user, among the highest of any genre. The fastest revenue lever is making the rewarded economy work harder. The model changes two things and adds no forced ad load:

- **Scale the reward to the player's income.** A reward worth "30 minutes of your current income/sec" stays meaningful from Bank 1 to the late game, where a flat reward becomes trivial as numbers grow. (Weber–Fechner: players perceive ratios, not absolutes.)
- **Resequence placement.** Showing an interstitial after a rewarded offer lifts ad ARPDAU by about +23% in industry tests, with no added frustration.

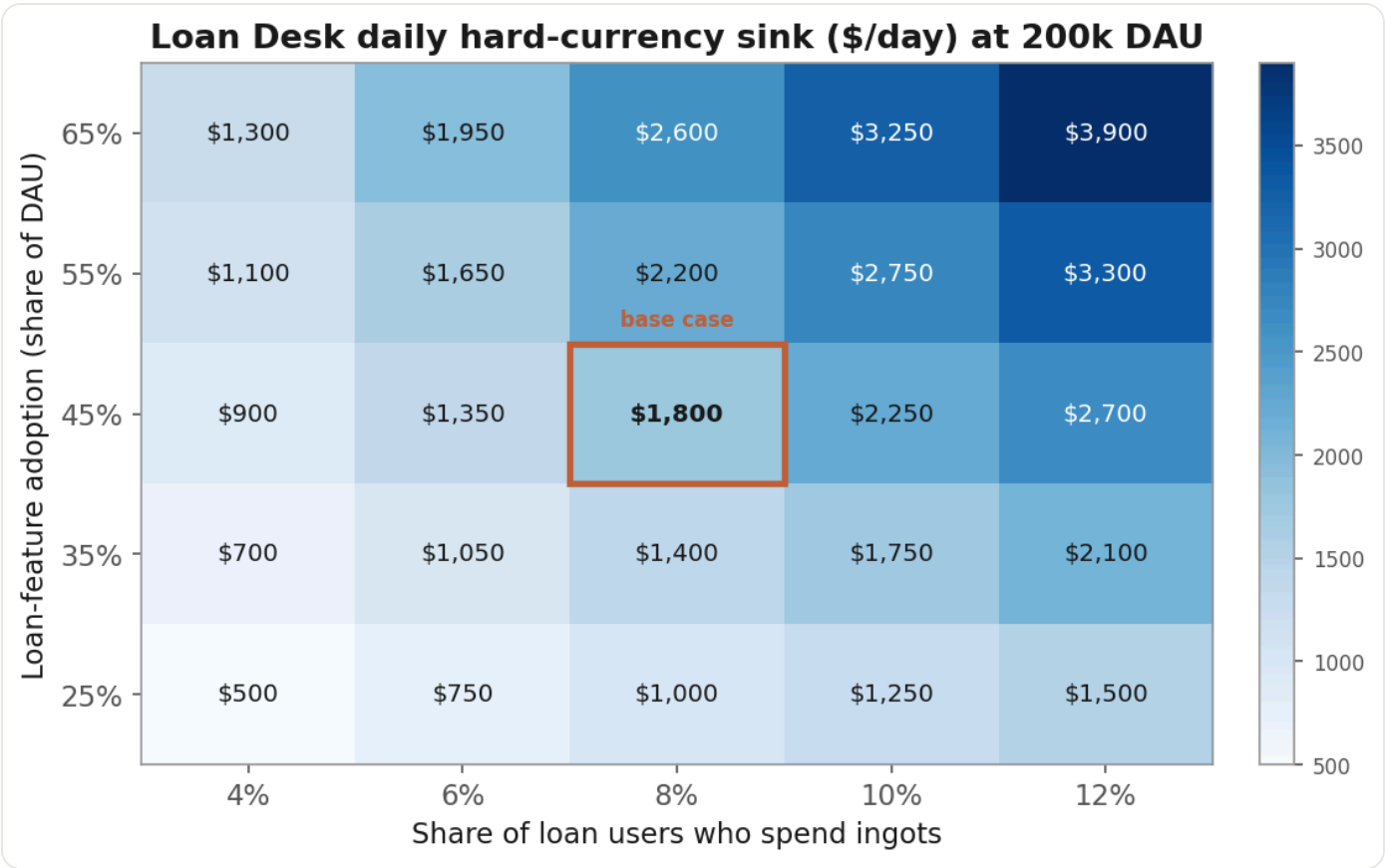
METRIC (200K DAU)	BASELINE	TUNED	Δ
Rewarded views / watcher / day	6.0	8.5	+41.7%
Daily rewarded ad revenue	~\$9,108	~\$12,903	+\$3,795
Incremental rewarded ARPDAU	\$0.0455	\$0.0645	+\$0.019

The 6.0 → 8.5 views figure is a modelling assumption, not a result: better-sized rewards raise willingness to watch. Only the +23% resequencing effect is sourced. Together they give **+41.7% daily rewarded-ad revenue from the same players.**

Combined ARPDau and sensitivity

LEVER	INCREMENTAL ARPDau
Baseline	\$0.180
+ Loan Desk (A)	+\$0.009
+ Ad tuning (B)	+\$0.019
Projected new ARPDau	\$0.208 (+15.5%)

The Loan Desk figure counts **only** the hard-currency sink. It does not count the retention value of giving late-game players something to do, which is the feature's main purpose and would raise the effect through higher D7+. The live grid shows the sink is robust: about **\$1,800/day** at the base case, spanning **\$500–\$3,900/day** across reasonable swings in adoption and spender share.



Daily hard-currency sink across adoption × spender share; the base case is outlined.

The A/B test plan

Every lever ships as an A/B test, not on the model's projection alone:

- **Design.** Assign at install (new players only). Split by platform, since iOS and Android behave differently. Test 4–5 variants of reward size and loan multiplier, not a single on/off.
- **Primary KPIs.** Evaluate retention and ARPU together, because tightening an economy often trades one for the other. Require $\geq 3\%$ ARPDAU movement before acting.
- **Guardrails.** Run the full 14-day retention window. Do not stack two tests on the same system. Watch the cash-balance distribution for inflation.
- **Decision rule.** Ship if D7 is up or flat and ARPDAU improves by $\geq 3\%$. Iterate if results are mixed. Stop if D1 drops or the sink triggers progression complaints.

How the model is built

Change one input and every dependent value recalculates. Every value is a single linked formula off a **Parameters** tab (400+ of them), with no hard-coded results, across seven tabs (Parameters, Baseline Economy, Loan Desk, Ad Tuning, KPI Impact, Sources). Change the cost-growth rate or eCPM once and the whole model rebalances. To explore it, open the workbook and change any blue cell.

Data and assumptions. Kolibri does not publish per-level cost tables or live KPIs, so every core-loop number here is a labelled **assumption**, calibrated to public third-party estimates (AppQuantum / Sensor Tower) and standard idle-game math (Pecorella, GDC). The contribution is the method, not the specific figures. Swapping in Kolibri's real telemetry would make the model production-grade without changing its structure. None of the figures are Kolibri's internal data.

What it took

Source/sink balancing

Master-parameter modelling

Excel (400+ formulas)

ROI & monetisation math

Sensitivity analysis

A/B test design

Retention & ARPDAU KPIs

Player segmentation

F2P benchmarking

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